



Grey quantum group decision and its application in meteorological disaster emergency

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ARTICLE INFO

Keywords:

Multi-attribute group decision making
Prospect theory
Grey numbers
2-additive Choquet integral
Quantum probability theory

ABSTRACT

In the context of sudden meteorological disasters, information is often uncertain, the attributes used to assess the disaster state are interrelated, and the opinions of decision makers may influence one another. Considering these problems, this paper proposes a novel quantum group information fusion model with grey numbers. Firstly, the 2-additive Choquet integral is adopted to reflect the correlations among attributes and determine attributes' weights. Secondly, considering the limited rationality of decision-makers during the disaster relief, the prospect theory is used to construct the performance value of rescue alternatives. Thirdly, a quantum-like Bayesian network (QLBN) is constructed to reflect interference effects among decision-makers' opinions. Grey relational degree is used to determine decision-makers' weights as the first layer of the QLBN. Based on the Bayesian principle, quantum probabilities of alternatives are calculated, and then alternatives are ranked by them. Finally, a case of evaluating meteorological disaster emergency plans verifies the effectiveness and superiority of this method. The robustness and effectiveness of the proposed method are further verified by comparative analysis, sensitivity analysis and simulation analysis.

1. Introduction

The research on multi-attribute group decision-making (MAGDM) is advancing at a rapid pace in the discipline of modern decision science (Rajput, Beg, & Kumar, 2024; Wan, Zou, Dong, & Gao, 2025). This research field is crucial, especially in addressing emergencies such as meteorological disasters. For example, during natural disasters like floods, typhoons, and other emergencies, effective MAGDM models are essential for making rescue decisions (Fang et al., 2023). There exists significant uncertainty in information during meteorological disasters, which makes it necessary to represent attribute values in various forms, such as crisp numbers (Chao, Kou, Peng, & Viedma, 2021), fuzzy terms (Deveci et al., 2023; Dong & Wan, 2024), linguistic terms (Büyüközkan, Uztürk, & Maden, 2023; Ma, Qin, Martínez, & Pedrycz, 2023) or grey numbers (Liu, Sheng, & Wang, 2023a; Yan, Liu, Liu, & Wu, 2015). In meteorological disaster decision-making, the urgency and unpredictability of time often present significant challenges in information gathering. Obtaining precise data can be very difficult in contexts with high uncertainty, such as in disaster emergency management. For instance, when assessing the number of injured individuals in a disaster, it may not be possible to

collect the count precisely. It could be estimated to be within a range like [1000, 2000]. While grey numbers and intervals share a similar mathematical form, they differ in modeling intent. Intervals primarily reflect numerical imprecision, whereas interval grey numbers, rooted in grey system theory, emphasize information incompleteness. Unlike intervals, grey numbers incorporate the concept of greyness, which indicates how much of the information is unknown. This feature makes them better suited for scenarios like disaster decision-making, where data are often incomplete, uncertain, and derived from heterogeneous sources. Therefore, the use of interval grey numbers is particularly effective when dealing with problems involving high uncertainty because it allows decision-makers to analyze and make decisions in situations where precise data is lacking (Kang & Dai, 2023; Oztaysi, 2014).

When dealing with the MAGDM situation, it is crucial to determine the weights assigned to different attributes, researchers have developed various methods to solve this problem (Zayat, Kilic, Yalcin, Zaim, & Delen, 2023). Commonly used approaches include the Analytic Hierarchy Process (AHP), Entropy Weight Method, and TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) (Shi, Xie, & Li, 2023; Sun, Zhou, Yang, & Huang, 2023b; Yang, Qi, & Xie, 2024). Although

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these methods are widely used, they often exhibit limitations when dealing with correlations or interactions among attributes, especially in mitigating the impact of subjective preferences on decision group members. While, for reflecting the interactions among attributes, the Choquet integral method is developed to assess various attributes' weights. Introducing the Choquet integral into the framework of fuzzy sets is a practical decision-making approach (Garg, 2024), which effectively maintains individual attributes' independence (Zhao, Ju, Martínez, Dong, & Shan, 2022). For instance, in the q-rung fuzzy environment, the application of the Choquet integral provides a novel perspective for the development and optimization of mobile healthcare applications (Tang et al., 2022). Furthermore, through the application of the Choquet integral, the suitability of the Intuitionistic Fuzzy Set theory in complex decision environments has been significantly enhanced (Divsalar, Ahmadi, Ebrahimi, & Ishizaka, 2022). By incorporating the Choquet integral into the image-based fuzzy VIKOR-TOPSIS method (Singh & Kumar, 2023) and the extended Exp-TODIM method, which combines Z-numbers and Wasserstein distance, addressing uncertainty and complexity (Sun, Yang, Cai, Wei, & Mo, 2023a). Some scholars have also explored the application of the Choquet integral and proposed the singleton Choquet-type set-valued integral based on the Aumann integral (De Li, Mesiar, & Pap, 2023). The Choquet integral plays a crucial role in assessing the correlation and importance of features (Mohammed et al., 2022; Yalcin, Kilic, & Delen, 2022). Therefore, it is evident that the Choquet integral plays a significant role in handling interactions among attributes, promoting a higher degree of independence among them.

Quantum Probability Theory (QPT) (Busmeyer & Bruza, 2012) as an inferential paradigm that overcomes the limitations of classical probability theory has shown considerable theoretical value and promising application potential in MAGDM in recent years (Liu, Liu, Jiang, & Han, 2023b; Xu, Yan, Pérez, & Cabrerizo, 2025c; Zhang, Zhu, & Tong, 2024). Traditional group decision-making models often assume that the opinions of decision-makers are independent and that their preferences remain stable (Zou, Wan, Dong, & Martínez, 2023). However, in practical decision-making processes, decision-makers frequently exhibit significant mutual interference, such as information sharing, emotional contagion, cognitive biases, and strategic behavior, all of which result in uncertainty and dynamic evolution in individual judgments (Wu, Liu, Qin, Wang, & Zhou, 2021; Wu, Liu, Zhou, Tao, & Qin, 2022; Xu, Yan, & Li, 2025b). By introducing core concepts such as superposition, interference effects, and non-commutativity, QPT enables the modeling of complex cognitive mechanisms and evolving preferences at the psychological level (Han & Liu, 2023; Tan, Zhu, & Wu, 2022). In recent years, numerous studies have sought to incorporate QPT into group decision-making models to enhance their adaptability and capacity for behavioral interpretation. For example, Wang, Liu, Li, Teng, and Pedrycz (2024) constructed a trust evolution network based on quantum theory, considering the interference effects of trust paths during trust transmission. Yan, Dong, Wan, and Gao (2024a) constructed a new decision-making model based on QPT to derive the weights of decision-makers and the ranking of alternatives. Yan, Xu, Gong, and Herrera-Viedma (2025) proposed a group decision-making model that combines intuitionistic fuzzy sets with QLBN, providing a practical solution for dynamic decision-making in the context of sudden-onset disasters. Zheng and Liu (2025) pointed out that existing HFACS-based methods lack the ability to quantify interference effects, and the QPT offers a novel perspective to model expert interference.

During the decision-making process, it is common for decision-makers to exhibit irrational behaviour, such as being averse to taking risks when presented with potential benefits and being inclined to take risks when faced with potential losses. Meanwhile, prospect theory effectively explains such irrational behaviour (Daniel & Amos, 2013). The prospect theory has been widely applied due to its relevance in various decision-making scenarios. For example, it has shown excellent performance in an enhanced fuzzy multi-attribute decision-making method for assessing agricultural robots (Brainy et al., 2023). Additionally, a

three-way decision model, rooted in the prospect-regret theory, efficiently tackles the problem of evaluating attribute weights in imperfect multi-scale decision information systems (Huang, Zhan, Xu, & Fujita, 2023). The effectiveness of a three-way decision model based on cumulative prospect theory has been proved in taking into account physicians' risk attitudes while making ambiguous judgements (Zhao, Wang, Meng, & Liao, 2023). In addition, the integration of prospect theory and probabilistic linguistic term sets within a three-way decision framework (PL-PT-TWD), along with the introduction of novel distance measures and satisfaction functions, has effectively facilitated the application of prospect theory in multi-attribute decision making (Wang, Zhan, & Zhang, 2023).

The motivation of this paper is as follows:

Motivation 1: In multi-attribute decision-making problems, there often exist complex dependencies and interactions among attributes. However, traditional weighting methods are generally based on the assumption that attributes are mutually independent, thus failing to account for their synergistic effects. The 2-additive Choquet integral provides a systematic approach to modeling both first-order and second-order interactions among attributes.

Motivation 2: Traditional MAGDM models typically aggregate individual opinions through information integration or weighted averaging, implicitly assuming that decision makers are independent and uninfluenced by others. In practice, however, group decision-making processes often involve information exchange, preference convergence, and cognitive linkage, leading to interference effects among decision makers. The quantum probability theory, through the construction of QLBN, offers a thought to model the complex cognitive structures and to effectively capture the interferences among decision makers.

Expanding on prior research, this paper proposes an innovative grey quantum group decision-making model aimed at improving the precision and efficiency of group decision processes. This model uses grey number to fuse information and capture the uncertainty in decision effectively. Additionally, it utilizes the 2-additive Choquet integral method to calculate attribute weights, reducing the impact of subjective preferences. Moreover, the model integrates quantum probability and prospect theory to address interactions and entanglement among decision-makers. This approach enables more precise and comprehensive analyses in complex decision environments.

The contributions of this paper are as follows:

- 1) A novel multi-attribute quantum group decision-making model considering information fusion is proposed, which accurately captures the interference effects on decision-makers' perspectives, thereby aligning it closely with real decision processes.
- 2) Considering the interaction effects among attributes, this paper determines the optimal attributes' weights based on the 2-additive Choquet integral.
- 3) Using grey relational degrees to measure the correlation between individual evaluation and overall evaluation, establishing the weights of decision-makers in the first layer of the BN.
- 4) Considering the finite rational behavior of decision makers, conditional probabilities in QLBN are calculated based on prospect theory.

The structure of this paper is organized as follows: Section 2 reviews the grey number, 2-additive Choquet integral, prospect theory, and the quantum probability theory. Section 3 proposes the grey multi-attribute quantum group decision-making model (GMAQGDM). This model integrates the 2-additive Choquet integral, prospect theory, quantum probability, and delineates the decision-making steps of GMAQGDM. Section 4 applies the model to an emergency rescue scenario in meteorological disasters, verifying its stability and reliability through sensitivity analysis, comparative analysis and simulation analysis. Finally, Section 5 provides a concise overview of the primary discoveries and contributions, as well as highlights potential avenues for further research.

2. Preliminaries

This section will briefly review several vital concepts and theories, including the grey numbers, 2-additive Choquet Integral, prospect theory, and quantum probability.

2.1. Grey numbers

Definition 1. (Yan et al., 2015) A grey number refers to a real number characterized by a known lower and upper bound, while the precise position of the value within this interval remains indeterminate.

Definition 2. (Lin & Liu, 2006) A grey number $\otimes \in [a, b] (a < b)$ is defined on domain Ω and $u(\otimes)$ represents the measure of the grey number $\otimes$, if $\otimes$ is a continuous grey number, $\hat{\otimes} = \frac{1}{2}(a + b)$ is the core of the grey number $\otimes$, $g^0(\otimes) = \frac{u(\otimes)}{u(\Omega)}$ is the grey degree of the grey number, and $\hat{\otimes}_{(g^0)}$ is the simplified form of the grey number.

The basic operations for grey numbers are as follows:

$$\hat{\otimes}_{1(g^0_1)} + \hat{\otimes}_{2(g^0_2)} = (\hat{\otimes}_1 + \hat{\otimes}_2)_{(g^0_1 \vee g^0_2)}. \quad (1)$$

$$\hat{\otimes}_{1(g^0_1)} - \hat{\otimes}_{2(g^0_2)} = (\hat{\otimes}_1 - \hat{\otimes}_2)_{(g^0_1 \vee g^0_2)}. \quad (2)$$

$$\hat{\otimes}_{1(g^0_1)} \times \hat{\otimes}_{2(g^0_2)} = (\hat{\otimes}_1 \times \hat{\otimes}_2)_{(g^0_1 \vee g^0_2)}. \quad (3)$$

$$\hat{\otimes}_{1(g^0_1)} / \hat{\otimes}_{2(g^0_2)} = (\hat{\otimes}_1 / \hat{\otimes}_2)_{(g^0_1 \vee g^0_2)}. \quad (4)$$

$$k \cdot \hat{\otimes}_{(g^0)} = (k \cdot \hat{\otimes})_{(g^0)}. \quad (5)$$

Definition 3. (Liu, Tao, Xie, & Yang, 2015) There are two standard interval grey numbers $\otimes_1 \in [a_1, b_1] \subset [0, 1]$ and $\otimes_2 \in [a_2, b_2] \subset [0, 1]$, the distance between $\otimes_1$ and $\otimes_2$ is as:

$$d(\otimes_1, \otimes_2) = \frac{1}{2}(|a_1 - a_2| + |b_1 - b_2|). \quad (6)$$

2.2. Mahalanobis–Taguchi system and the 2-additive Choquet integral

Definition 4. (Grabisch, 1997; Michio, 1974) Let $C = \{c_1, c_2, \dots, c_n\}$ be a non-empty finite set, $F(C)$ be the power set of C , and u be a fuzzy measure defined on $(C, F(C))$, $|C| = n$, $\forall B \in F(C)$, $|B| \geq 2$, the interaction index can be defined as:

$$I_B = \sum_{k=0}^{n-|B|} \xi_k^{|B|} \sum_{K \subseteq C \setminus B, |K|=k} \sum_{L \subseteq B} (-1)^{|B|-|L|} \mu_{KL}. \quad (7)$$

Grabisch offers a k -additive fuzzy measure using the Möbius transform. Additionally, Grabisch presents a 2-additive fuzzy measure derived from this, which significantly decreases the complexity of the fuzzy measure. The set K is defined as the complement of set B in set C . In other words, K is a subset of the elements in C that are not in B . Let L be a subset of B , denoted as $L \subseteq B$. The term $\xi_k^{|B|}$ may be defined in the following manner:

$$\xi_k^{|B|} = \frac{(n-k-|B|)!k!}{(n-|B|+1)!}. \quad (8)$$

If $B = \{c_i\}$, I_i is the Shapley value of the attribute c_i , which can be interpreted as the average importance of the attribute c_i relative to other attributes. If $B = \{c_i, c_j\}$, I_{ij} is the interaction index between attributes c_i and c_j . If $I_{ij} > 0$, then c_i and c_j are mutually reinforcing; if $I_{ij} < 0$, then c_i and c_j are mutually suppression, if $I_{ij} = 0$, then c_i and c_j are independent of each other.

Given the unique attributes of the Mahalanobis–Taguchi system (MTS) approach in handling uncertainty and fuzziness, the method entails the following steps in assigning attribute weights through the fuzzy measures and the 2-additive Choquet integral (Chang, Cheng, & Liu, 2016).

There are two distinct sample sets $G = \{y_h | h = 1, 2, \dots, m\}$ and $\hat{G} = \{\hat{y}'_h | h = 1, 2, \dots, l\}$, and the sample data matrix is $Y = [y_h(c_j)]_{m \times n}$ and $\hat{Y} = [\hat{y}'_h(c_j)]_{l \times n}$ respectively.

According to the sample data matrix $Y = [y_h(c_j)]_{m \times n}$, $y_h(c_j)$ represents the value corresponding to c_j in the sample data. The mean value and standard deviation of each attribute are calculated.

$$\mu(c_j) = \frac{1}{m} \sum_{h=1}^m y_h(c_j), j = 1, 2, \dots, n. \quad (9)$$

$$s(c_j) = \sqrt{\frac{1}{m-1} \sum_{h=1}^m (y_h(c_j) - \mu(c_j))^2}, j = 1, 2, \dots, n. \quad (10)$$

Utilizing the mean and standard deviation, standardize the sample data matrix as follows:

$$\hat{v}_h(y_h(c_j)) = \frac{y_h(c_j) - \mu(c_j)}{s(c_j)}, h = 1, 2, \dots, m, j = 1, 2, \dots, n. \quad (11)$$

The normalized sample data matrix $Q = [\hat{v}_h(y_h(c_j))]_{m \times n}$ can be obtained, and the interrelation coefficient matrix between attributes can then be calculated:

$$R_B = \frac{1}{m-1} \sum_{h=1}^m (Q_B^{(h)})^T Q_B^{(h)}. \quad (12)$$

where, R_B is $|B| \times |B|$ matrix, $|B|$ is the number of attributes in the attribute set B . $Q_B^{(h)}$ is the h -th datum represented by the attribute set B .

Definition 5. (Genichi, Gen'ichi, & Rajesh, 2002) Mahalanobis distance is a measure of similarity or difference between samples, calculated as follows:

$$d_B^{(h)} = \frac{1}{n} Q_B^{(h)} R_B^{-1} (Q_B^{(h)})^T, h = 1, 2, \dots, l. \quad (13)$$

The significance level is :

$$\eta_B = -10 \log_{10} \left[\frac{1}{l} \sum_{h=1}^l \frac{1}{d_B^{(h)}} \right]. \quad (14)$$

As the square root of η_B in MTS grows, the significance of attribute set B in the classification process also increases.

Definition 6. (Michel & Christophe, 2010) The function f is a non-negative real-valued function defined on the set C . The set C represents a collection of attributes c_j , while the set C^* represents the complement of the set C . The definition of this is based on the Shapley values of each attribute and the interaction indices between attributes, which are calculated using the 2-additive Choquet integral formula.

$$C_\mu(f) = \sum_{j=1}^n I_j f(c_j) - \frac{1}{2} \sum_{\{c_j, c_p\} \subset C} I_{jp} |f(c_j) - f(c_p)|. \quad (15)$$

In addition, the single-attribute indicator I_j and the attribute-interaction indicator I_{jp} must comply with the criteria of monotonicity and regularity, which are described as follows.

$$-2I_j \leq \sum_{c_p \in C^* \setminus \{c_j\}} I_{jp} - \sum_{c_p \in C} I_{jp} \leq 2I_j, j = 1, 2, \dots, n. \quad (16)$$

2.3. Prospect theory

Prospect theory (Kahneman & Tversky, 1979) was proposed in response to the significant divergence between actual human behavior and the predictions of expected utility theory in real-world decision-making. It effectively captures individuals' irrational risk preferences and better reflects the bounded rationality of decision-makers under uncertainty. Prospect value is defined by both its value function and its probability weight function. If the value of the index is larger than the reference

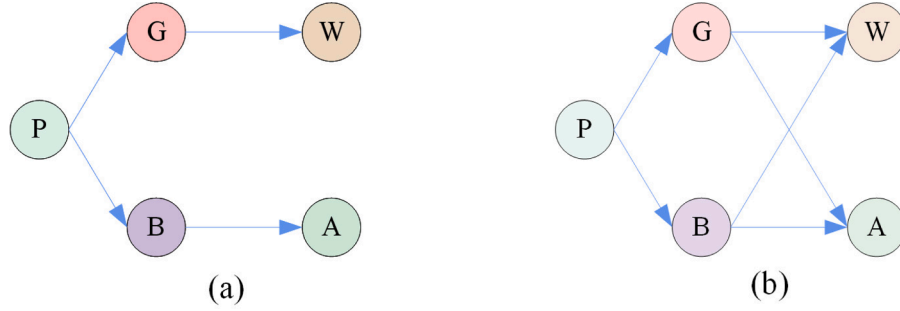


Fig. 1. The graphical representation of a Markov chain's paths.

point, the decision maker will experience benefits and exhibit risk aversion. When the index value is lower than the reference point, the decision maker encounters a situation of potential loss and chooses to take risks. The value function is defined in the following manner:

$$v(\hat{x}) = \begin{cases} (\hat{x} - m)^\alpha, & \hat{x} > m; \\ -\theta(-\hat{x} + m)^\beta, & \hat{x} < m. \end{cases} \quad (17)$$

where, $\hat{x}$ is the result, m is the reference point, the coefficients α ($0 < \alpha < 1$) and β ($0 < \beta < 1$) represent the risk attitude of decision-makers; higher values of α and β indicate a greater propensity for risk-taking. The coefficient θ , known as the loss aversion coefficient, quantifies sensitivity to losses; a value of θ greater than 1 suggests that decision-makers are particularly loss-averse. The variable v denotes the decision value, which is a numerical representation of the overall assessment of an option within the decision-making process. This framework allows for a structured quantification of risk preferences and loss sensitivity, which are crucial for modeling decision-maker behaviors in various scenarios.

In risk decision-making, people's perception of probabilities exhibits nonlinear characteristics, which means that their subjective perception of probabilities is not directly proportional to actual probabilities. Probability weighting functions are used to describe this perceptual bias and are defined as follows:

$$\pi(p) = \begin{cases} \frac{p^{\hat{\gamma}}}{(p^{\hat{\gamma}} + (1-p)^{\hat{\gamma}})^{1/\hat{\gamma}}}, & \text{for gains;} \\ \frac{p^{\hat{\delta}}}{(p^{\hat{\delta}} + (1-p)^{\hat{\delta}})^{1/\hat{\delta}}}, & \text{for losses.} \end{cases} \quad (18)$$

where p is the actual probability, Parameters $\hat{\gamma}$ and $\hat{\delta}$ represent the value facing gain and loss respectively, and mainly control the curvature of the foreground weight function curve. According to experiments, it has been shown that when the parameter is set to $\alpha = \hat{\beta} = 0.88$, $\theta = 2.25$, $\hat{\gamma} = 0.61$, $\hat{\delta} = 0.69$, it aligns more closely with empirical data (Krohling & de Souza, 2012). The prospective value is defined as:

$$V(\hat{x}) = \pi(p) \cdot v(\hat{x}). \quad (19)$$

2.4. Decision-making in quantum probability theory

Quantum probability theory is being increasingly used in decision theory, representing a distinct departure from conventional decision theory. Traditional decision theory posits that humans systematically assess all available options and then make optimum decisions based on these assessments. Conversely, quantum probability theory provides an alternative viewpoint that incorporates non-classical probability calculations and examines how humans make judgements when confronted with ambiguity and complexity (Xu, Yan, & Cabrerizo, 2025a). Fig. 1 presents a Markov chain route diagram that models the progression of decision paths, illustrating the temporal evolution of states within the decision-making process.

In Fig. 1(a), in accordance with the Law of Total Probability within classical probability theory, $P(A)$ can be determined through the application of the principle of total probability.

$$P(A) = P(G)P(A|G) + P(B)P(A|B). \quad (20)$$

In quantum theory, the transition probability amplitudes between states may be determined by calculating the inner product of these two states (Yan, Xu, Gong, & Herrera-Viedma, 2024b). For instance, in Fig. 1(b), the state transition from $P \rightarrow G$ can be expressed using Dirac notation $\langle G|P \rangle$. It is important to note that these amplitudes are complex numbers, not real numbers. The classical probability values associated with these amplitudes are equal to the square modulus of the amplitudes, denoted as $|\langle G|P \rangle|^2$ and $|\langle A|G \rangle|^2$. In this explicit choice process, the outcomes derived from quantum theory correspond with those of classical theory. Nevertheless, the computation of probabilities for $P \rightarrow A$ is very intricate and may be articulated using a technique akin to the law of total probability. on facilitate comparison, one may apply an Euler transformation on $\langle G|P \rangle \cdot \langle A|G \rangle$ and $\langle B|P \rangle \cdot \langle A|B \rangle$ in the equation in order to expose the disparities in probability estimates between these two alternatives:

$$\begin{aligned} p(A) &= |\langle A|P \rangle|^2 = |\langle G|P \rangle \cdot \langle A|G \rangle + \langle B|P \rangle \cdot \langle A|B \rangle|^2 \\ &= \left(\sqrt{p(G)p(A|G)} \cdot e^{i\alpha} + \sqrt{p(B)p(A|B)} \cdot e^{i\beta} \right) \\ &\quad \cdot \left(\sqrt{p(G)p(A|G)} \cdot e^{-i\alpha} + \sqrt{p(B)p(A|B)} \cdot e^{-i\beta} \right) \\ &= p(G)p(A|G) + p(B)p(A|B) + \\ &\quad 2\sqrt{p(G)p(A|G)p(B)p(A|B)} \cdot \cos(\alpha - \beta). \end{aligned} \quad (21)$$

In quantum probability, the formula includes an additional element called the interference term, which is $2\sqrt{p(G)p(A|G)p(B)p(A|B)} \cdot \cos(\alpha - \beta)$. This term is not included in classical probability theory. The interference term is a result of the interaction between two distinct probability amplitudes. The interference phenomena seen in quantum theory is distinct and arises from the properties of superposition and interaction of quantum states, which are absent in conventional probability theory. The inclusion of the interference term is a crucial component that distinguishes quantum decision processes from conventional probability choices. It has the capability to enhance comprehension of the behaviour of quantum systems.

3. Multi-attribute quantum group decision-making model based on grey number information

The method framework mainly includes the following four processes:

- (1) Collection and processing of evaluation information. The evaluation information of decision makers on all alternatives under different attributes was collected and expressed by interval grey number to describe the information incompleteness. On this basis, the normalization process is carried out to eliminate the interference of dimension difference on subsequent analysis.
- (2) Determination of attributes' weights. In order to describe the first-order and second-order interactions between attributes, the 2-additive Choquet integral method based on Maten system is introduced in this paper. Maten system provides the structural basis for the construction of fuzzy measures and helps to ensure the theoretical consistency of integral forms under non-additive measures.

Table 1
Summarization of mathematical symbols.

Symbol	Description	Symbol	Description
D	Set of decision-makers $\{d_1, \dots, d_s\}$	X	Set of alternatives $\{x_1, \dots, x_m\}$
C	Set of attributes $\{c_1, \dots, c_n\}$	$a_{ij}^k(\otimes)$	Interval grey number $[a_{ij}^{kL}, a_{ij}^{kU}]$
ω_j^k	Weight of attribute c_j by decision-maker d_k	$F(C)$	Power set of attribute set C
B	Attribute subset, $B \in F(C)$	λ^0	Subjective weight vector of decision-makers
$\mu(c_j^k)$	Mean of c_j by decision-maker d_k	$s(c_j^k)$	Standard deviation of c_j by d_k
$\hat{v}_{ij}^k(\cdot)$	Normalized evaluation value	R_B^k	Correlation matrix of attributes in subset B
η_B	Importance degree of attribute subset B	I_j	Importance degree of attribute c_j
$C_\mu^k(c_j)$	Choquet integral value for c_j by d_k	$v(\cdot)$	Value function in prospect theory
$\pi_{ij}^k(\cdot)$	Weighting function in prospect theory	V_i^k	Prospect value of x_i by decision-maker d_k
$\varepsilon_{\rho k}$	Grey absolute relational coefficient	$\hat{R}^k$	Behavior sequence of decision-maker d_k
$\hat{R}^+$	Ideal reference sequence	$\tilde{\lambda}_k$	Aggregated weight of decision-maker d_k
λ_k^0	Subjective weight of d_k	λ_k	Objective weight of d_k
g	Fusion coefficient for subjective-objective weights	$p(d_k)$	Prior probability in quantum Bayesian network
$p(x_i d_k)$	Conditional probability in QLBN	$p(x_i)$	Quantum probability of alternative x_i

- (3) Determination of decision-makers' weight. The grey absolute correlation degree model is used to calculate the correlation degree and determine the objective weight. Combined with subjective weighting, a linear weighted fusion mechanism is used to obtain the final weight that takes into account cognitive reliability and subjective judgment, and provides prior distribution support for quantum Bayes model.
- (4) Construction of quantum-like Bayesian framework and quantum probability calculation. A two-layer quantum-like Bayesian model is constructed. The first layer is the prior probability, which is the comprehensive weight of each decision maker. The second layer is conditional probability based on prospect theory transformation, that is, the prospect value of each decision maker's alternative scheme is calculated, the preference vector is constructed, and the positive distribution vector is transformed by exponential mapping. Finally, the quantum probability values of each alternative scheme are calculated and sorted.

In order to facilitate understanding, the main mathematical symbols in this paper are summarized in Table 1.

3.1. Collection and processing of evaluation information

3.1.1. Evaluation information collection

The initial information matrix $A^{(k)} = (a_{ij}^k(\otimes))_{m \times n}$ of decision-maker d_k ($k = 1, 2, \dots, s$) is:

$$A^{(k)} = \begin{bmatrix} a_{11}^k(\otimes) & a_{12}^k(\otimes) & \dots & a_{1n}^k(\otimes) \\ a_{21}^k(\otimes) & a_{22}^k(\otimes) & \dots & a_{2n}^k(\otimes) \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}^k(\otimes) & a_{m2}^k(\otimes) & \dots & a_{mn}^k(\otimes) \end{bmatrix}. \quad (22)$$

3.1.2. Normalize the initial information matrix

Because of the different nature of each evaluation index, it usually has different dimensions and orders of magnitude. In order to eliminate dimensions and units and facilitate the aggregation of index values, the index values need to be normalized. According to the defined value domain, the gray number is converted into kernel and gray forms. The normalization form $\tilde{a}_{ij}^k(\otimes_{(g^0)})$ is defined as follows:

$$\tilde{a}_{ij}^k(\otimes_{(g^0)}) = \begin{cases} \frac{a_{ij}^k(\otimes_{(g^0)}) - \min_i a_{ij}^k(\otimes_{(g^0)})}{\max_i a_{ij}^k(\otimes_{(g^0)}) - \min_i a_{ij}^k(\otimes_{(g^0)})}, & \text{if } c_j \text{ is a benefit attribute} \\ \frac{\max_i a_{ij}^k(\otimes_{(g^0)}) - a_{ij}^k(\otimes_{(g^0)})}{\max_i a_{ij}^k(\otimes_{(g^0)}) - \min_i a_{ij}^k(\otimes_{(g^0)})}, & \text{if } c_j \text{ is a cost attribute} \end{cases}$$

$i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s.$

(23)

3.2. The comprehensive prospective values of alternatives for each decision-maker

3.2.1. Determination of attribute weights considering the correlation effects

In MAGDM problems, attributes exhibit interdependencies and correlations. To account for such interactions, the 2-additive Choquet integral based on the measure of MTS is employed to determine attributes' weights.

- 1) Calculate the mean and standard deviation for each attribute:

$$\mu(c_j^k) = \frac{1}{m} \sum_{i=1}^m a_{ij}^k(\otimes_{(g^0)}), j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (24)$$

$$s(c_j^k) = \sqrt{\frac{1}{m-1} \sum_{i=1}^m [a_{ij}^k(\otimes_{(g^0)}) - \mu(c_j^k)]^2}, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (25)$$

Standardize the sample data matrix using the mean and standard deviation:

$$\hat{v}_{ij}^k(\otimes_{(g^0)}) = \frac{a_{ij}^k(\otimes_{(g^0)}) - \mu(c_j^k)}{s(c_j^k)}, i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (26)$$

The standardized sample data matrix $Q_B^k = [\hat{v}_{ij}^k(\otimes_{(g^0)})]_{m \times n}$ is obtained.

- 2) Calculate the interrelation coefficient matrix between attributes:

$$R_B^k = \frac{1}{n-1} \sum_{j=1}^n (Q_B^{kj})^T Q_B^{kj}, k = 1, 2, \dots, s. \quad (27)$$

where, Q_B^{kj} is the j -th sample data in the attribute set based on decision-maker d_k .

- 3) Calculate the Mahalanobis distance:

$$d_B^{(j)} = \frac{1}{n} (Q_B^{kj})^T R_B^{-1} (Q_B^{kj}), j = 1, 2, \dots, l, l \leq n. \quad (28)$$

- 4) Calculate the importance level:

$$\eta_B = -10 \log_{10} \left[\frac{1}{l} \sum_{j=1}^l \frac{1}{d_B^{(j)}} \right]. \quad (29)$$

5) For MAGDM problems, it is argued that optimal decision-making outcomes are achieved by ensuring positive cooperation among the attributes within the attribute set. An optimization model has been constructed to solve for the global importance, denoted as η_j . And a series of ε variables have been introduced as slack variables. These variables are essential for quantifying and managing the uncertainty and adaptability

of the model. The model is formulated as:

$$\begin{aligned} & \max \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_{2^n - n - 1} \\ & \text{s.t.} \begin{cases} \eta_B - \sum_{(c_j) \in B} \eta_j \geq \varepsilon_p, p = 1, 2, \dots, (2^n - n - 1), \\ -1 \leq \varepsilon_p \leq 1, p = 1, 2, \dots, (2^n - n - 1), \\ 0 \leq \eta_j \leq 1, \sum_{j=1}^n \eta_j = 1, j = 1, 2, \dots, n. \end{cases} \end{aligned} \quad (30)$$

If $|C| = n$, then there are $(2^n - n - 1)$ properties set B . Calculate the Shapley value for an attribute:

$$I_j = \frac{\eta_j}{\sum_{j=1}^n \eta_j}, j = 1, 2, \dots, n. \quad (31)$$

6) The optimal interaction indices should ensure that each decision alternative can compete on a fair basis. Therefore, considering the interaction index's value range as constraint function, choose the maximum composite attribute values as objective function, for ensuring equal competition among decision alternatives. Construct the following multi-objective optimization model to determine the interaction indices I_{jp} .

$$\begin{aligned} & \max \sum_{j=1}^n |C_\mu(c_j)| \\ & \text{s.t.} \begin{cases} -2I_j \leq \sum_{(c_p) \in C^* \setminus \{c_j\}} I_{jp} - \sum_{(c_p) \in C} I_{jp} \leq 2I_j, j = 1, 2, \dots, n, p = 1, 2, \dots, n, \\ -1 \leq I_{jp} \leq 1, \{c_j, c_p\} \subseteq C. \end{cases} \end{aligned} \quad (32)$$

Theorem 1. The Model (32) has an optimal solution.

Proof. Firstly, since the goal function includes absolute values, which have continuous qualities, it may be inferred that the objective function is also continuous. Furthermore, because every element in both the information evaluation matrix and the Choquet integral has a limit, it can be concluded that the objective function likewise has a limit. The model's restrictions may be described as a closed set. Since the variables are limited to closed intervals, the set determined by these restrictions is naturally closed.

The Weierstrass approximation theorem (Chabert & Peruginelli, 2018) states that for a continuous function defined on a closed and limited set, it is guaranteed to reach its maximum and lowest values inside that set. Since the objective function of this model is both continuous and limited, and the constraints create a closed set, the feasible domain of this model is guaranteed to exist and has boundaries. This research validates the optimisation issue as a well-defined single objective model.

Based on the analysis provided, taking into account the continuous and limited nature of the objective function and the closed set established by the constraints, it is evident that model Eq. (32) has an optimum solution. □

7) Calculate the scores for each attribute based on the 2-additive Choquet integral.

$$C_\mu^k(c_j) = \sum_{i=1}^m I_i \bar{a}_{ij}^k(\hat{\otimes}_{(g^0)}) - \frac{1}{2} \sum_{\{c_i, c_q\} \subset C} I_{iq} [\bar{a}_{ij}^k(\hat{\otimes}_{(g^0)}) - \bar{a}_{iq}^k(\hat{\otimes}_{(g^0)})], \quad j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (33)$$

Finally, calculate attribute weight based on the 2-additive Choquet integral:

$$\omega_j^k = \frac{C_\mu^k(c_j)}{\sum_{j=1}^n C_\mu^k(c_j)}, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (34)$$

3.2.2. Comprehensive prospective values for alternatives under each decision-maker

To describe irrational behaviour in the decision-making process, this paper utilize prospect theory to construct alternative prospect values reflecting the decision-maker's irrational risk attitudes.

Using the attribute mean as the reference point, and it is denoted as:

$$\bar{a}_{ij}^k(\hat{\otimes}_{(g^0)}) = \frac{1}{m} \sum_{i=1}^m a_{ij}^k(\hat{\otimes}_{(g^0)}), j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (35)$$

Establish the value function and weighting function for prospect theory. The value function is denoted as:

$$v(a_{ij}^k(\hat{\otimes}_{(g^0)})) = \begin{cases} d(a_{ij}^k(\hat{\otimes}_{(g^0)}), \bar{a}_{ij}^k(\hat{\otimes}_{(g^0)}))^\alpha, a_{ij}^k(\hat{\otimes}_{(g^0)}) > \bar{a}_{ij}^k(\hat{\otimes}_{(g^0)}). \\ -\theta d(a_{ij}^k(\hat{\otimes}_{(g^0)}), \bar{a}_{ij}^k(\hat{\otimes}_{(g^0)}))^\beta, a_{ij}^k(\hat{\otimes}_{(g^0)}) < \bar{a}_{ij}^k(\hat{\otimes}_{(g^0)}). \end{cases} \quad i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (36)$$

To eliminate the influence of dimensions on calculations, normalize all prospect values uniformly according to the following formula:

$$\bar{v}(a_{ij}^k(\hat{\otimes}_{(g^0)})) = \frac{v(a_{ij}^k(\hat{\otimes}_{(g^0)}))}{\max_{i=1, \dots, m} |v(a_{ij}^k(\hat{\otimes}_{(g^0)}))|}, i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (37)$$

The weighting function is:

$$\pi_{ij}^k(\omega_j^k) = \begin{cases} \frac{(\omega_j^k)^{\hat{\gamma}}}{[(\omega_j^k)^{\hat{\gamma}} + (1 - \omega_j^k)^{\hat{\gamma}}]^{1/\hat{\gamma}}}, a_{ij}^k(\hat{\otimes}_{(g^0)}) > \bar{a}_{ij}^k(\hat{\otimes}_{(g^0)}). \\ \frac{(\omega_j^k)^{\hat{\delta}}}{[(\omega_j^k)^{\hat{\delta}} + (1 - \omega_j^k)^{\hat{\delta}}]^{1/\hat{\delta}}}, a_{ij}^k(\hat{\otimes}_{(g^0)}) < \bar{a}_{ij}^k(\hat{\otimes}_{(g^0)}). \end{cases} \quad i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s. \quad (38)$$

The comprehensive prospect value about alternative x_i by decision-maker d_k is obtained as:

$$V_i^k(\hat{\otimes}_{(g^0)}) = \sum_{j=1}^n \bar{v}(a_{ij}^k(\hat{\otimes}_{(g^0)})) \pi_{ij}^k(\omega_j^k), i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (39)$$

Moreover, $V^k(X)$ represents the prospect value of each alternative based on decision maker d_k , where $V^k(X) = (V^k(x_1), V^k(x_2), \dots, V^k(x_m))^T (k = 1, 2, \dots, s)$.

3.3. Determination of decision-makers' weights based on grey absolute relational degree

The concept of grey absolute relational degree is employed to quantify the degree of closeness between a reference sequence and a comparative sequence. This technique assesses the overall performance and benefits of each choice scheme by comparing the similarity between each reference sequence and the behaviour sequence. This technique not only simplifies the process of measuring decision-makers' preferences, but also emphasises the calculation of the level of correlation among decision-makers. This approach ensures that the assignment of weights to each member is more objective and rational (Liu et al., 2015). A decision-maker weight model is proposed based on this correlation. This model is incorporated as a probabilistic element in the first layer of the QLBN.

$$\hat{R}^k(\hat{\otimes}_{(g^0)}) = (\hat{r}_1^k(\hat{\otimes}_{(g^0)}), \hat{r}_2^k(\hat{\otimes}_{(g^0)}), \dots, \hat{r}_m^k(\hat{\otimes}_{(g^0)})), k = 1, 2, \dots, s. \quad (40)$$

Where,

$$\hat{r}_i^k(\hat{\otimes}_{(g^0)}) = \frac{1}{n} \sum_{j=1}^n \bar{a}_{ij}^k(\hat{\otimes}_{(g^0)}), i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (41)$$

The maximum value of the comprehensive evaluation of each alternative by all decision makers is calculated.

$$\hat{r}_i^+(\hat{\otimes}_{(g^0)}) = \max_{1 \leq k \leq s} \hat{r}_i^k(\hat{\otimes}_{(g^0)}), i = 1, 2, \dots, m. \quad (42)$$

The sequence being referenced is:

$$\hat{R}^+(\hat{\otimes}_{(g^0)}) = (r_1^+(\hat{\otimes}_{(g^0)}), r_2^+(\hat{\otimes}_{(g^0)}), \dots, r_m^+(\hat{\otimes}_{(g^0)})). \quad (43)$$

Therefore:

$$|s_k| = \left| \sum_{i=2}^{m-1} \hat{r}_i^k(\hat{\otimes}_{(g^0)}) + \frac{1}{2} \hat{r}_m^k(\hat{\otimes}_{(g^0)}) \right|, k = 1, 2, \dots, s. \quad (44)$$

$$|s_o| = \left| \sum_{i=2}^{m-1} \hat{r}_i^+(\hat{\otimes}_{(g^0)}) + \frac{1}{2} \hat{r}_m^+(\hat{\otimes}_{(g^0)}) \right|. \quad (45)$$

The difference between the reference sequence and the behavior sequence is defined as:

$$|s_k - s_o| = \left| \sum_{i=2}^{m-1} (\hat{r}_i^k(\hat{\otimes}_{(g^0)}) - \hat{r}_i^+(\hat{\otimes}_{(g^0)})) + \frac{1}{2} (\hat{r}_m^k(\hat{\otimes}_{(g^0)}) - \hat{r}_m^+(\hat{\otimes}_{(g^0)})) \right|, \quad k = 1, 2, \dots, s. \quad (46)$$

Calculate the grey absolute correlation:

$$\varepsilon_{ok} = \frac{1 + |s_k| + |s_o|}{1 + |s_k| + |s_o| + |s_o - s_k|}, k = 1, 2, \dots, s. \quad (47)$$

Theorem 2. The closer the behavior sequence $\hat{R}^k(\otimes)$ is to the reference sequence $\hat{R}^+(\otimes)$, the greater the correlation ε_{ok} between them.

Proof. Initially, the sequence $\hat{R}^k(\otimes)$, representative of averaged evaluations from decision makers, contrasts with $\hat{R}^+(\otimes)$, which is defined as the maximal average across all decision makers. These sequences are fundamental for computing the difference measure $|s_k - s_o|$. This measure quantifies the divergence of each decision maker's sequence from the reference sequence, as detailed in Eqs. (44)–(46).

The core of the theorem lies in the behavior of the grey absolute correlation ε_{ok} , as outlined in Eq. (47). This correlation increases inversely with the decrease in $|s_k - s_o|$. Designed to normalize the differences, ε_{ok} approaches unity as the alignment between $\hat{R}^k(\otimes)$ and $\hat{R}^+(\otimes)$ tightens, clearly demonstrating that diminished discrepancies correlate with augmented correlations.

Thus, the theorem is validated by showing that a decrease in $|\hat{s}_k - \hat{s}_o|$ directly leads to an increase in ε_{ok} . This relationship underscores a stronger correlation as the behavior sequence more closely approximates the reference sequence. This indicates that the closer the sequence $\hat{R}^k$ is to the reference sequence $\hat{R}^+$, the greater the relational grade ε_{ok} between them. $\square$

Finally, calculate the decision-maker's weight:

$$\lambda_k = \frac{\varepsilon_{ok}}{\sum_{k=1}^s \varepsilon_{ok}}, k = 1, 2, \dots, s. \quad (48)$$

Considering the subjective weight vector $\lambda_k^0 (k = 1, \dots, s)$ as a basis, the comprehensive weight of the decision group is calculated as follows, where the weight coefficient g is satisfying $0 \leq g \leq 1$.

$$\tilde{\lambda}_k = g\lambda_k^0 + (1 - g)\lambda_k, k = 1, 2, \dots, s. \quad (49)$$

3.4. Aggregation of alternatives' comprehensive evaluation values based on the quantum framework

Meteorological emergency response procedures are complex, ever-changing, and include a multitude of decision-makers, there are distinct preferences and assessments. In order to mitigate the impact of disasters quickly, accurate and fair decision-making is needed. As a result, our model was developed especially to meet the unique requirements of emergency management for meteorological disasters.

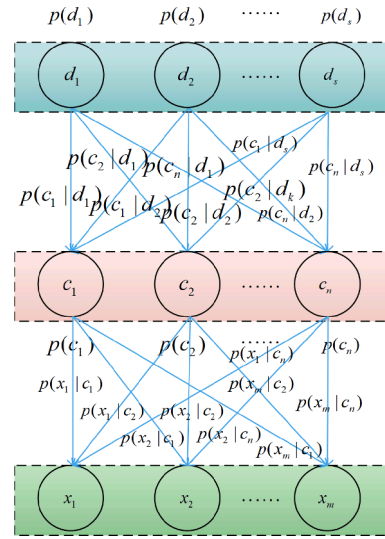


Fig. 2. Bayesian networks in the GMAQGDM problem of meteorological disaster.

In this paper, we adopt a quantum decision-making framework combined with a three-level BN to simulate the GMAQGDM process. The uppermost layer of this network delineates various decision-making groups, highlighting the influence of peer inputs and preferences on each decision-maker's choices. The intermediate layer maps the attributes relevant to MAGDM scenarios, and the lowest layer enumerates all potential rescue alternatives, thereby constituting the solutions layer. This model effectively depicts the interplay between different attributes and decision-makers. The first layer is identified as the team of rescue professionals, the second is meteorological elements. The third layer is feasible rescue strategies. Fig. 2 visually represents this comprehensive process.

Let the decision maker's weights be the probabilities of the first layer nodes (decision-makers) in the Bayesian network:

$$p(d_k) = \tilde{\lambda}_k, k = 1, 2, \dots, s. \quad (50)$$

Based on the prospect theory, the prospect value of each alternative for each decision-maker is converted into conditional probability. However, since the prospect value may contain negative values, direct use of probability modeling will lead to non-positive problems. Therefore, this paper adopts an exponential mapping based on the natural constant e to ensure that the calculated probability values are positive. The conditional probability is calculated as follows:

$$p(x_i|d_k) = \frac{e^{V_i^k}}{\sum_{k=1}^s e^{V_i^k}}, i = 1, 2, \dots, m, k = 1, 2, \dots, s. \quad (51)$$

In the framework of QPT, the probability score of a alternative x_i is calculated as the square magnitude of the aggregated amplitude probabilities across all feasible paths.

$$p(x_i) = \sum_{k=1}^s p(d_k)p(x_i|d_k) + 2 \sum_{k=1}^{s-1} \sum_{k'=k+1}^s \sqrt{p(d_k)p(x_i|d_k)p(d_{k'})p(x_i|d_{k'})} \cos(\theta_k - \theta_{k'}), \quad i = 1, 2, \dots, m. \quad (52)$$

Analyze the angular differences among different decision-makers to determine the probability of x_i , and then rank the alternatives based on the probability value of x_i .

3.5. GMAQGDM model decision steps

In order to improve the interpretability of the proposed model to the general reader, the key inputs and outputs of this model are as follows:

- 1) Model inputs include: a) an evaluation information matrix for each decision maker; b) Parameters required by prospect theory, including value function parameters and decision power function parameters.
- 2) The model output includes: a) the weight of the decision maker; b) The weight of the attributes of each decision maker; c) Quantum probabilities and final sorting results of alternatives.

The framework of the method proposed in this paper is shown in the Fig. 3, and the specific modeling steps are as follows:

Step 1: Collect evaluation information. The decision-maker d_k ($k = 1, 2, \dots, s$) assesses attributes c_j ($j = 1, 2, \dots, n$) for the alternative x_i ($i = 1, 2, \dots, m$), resulting in the matrix $A^{(k)} = (a_{ij}^k(\otimes))_{m \times n}$ ($k = 1, 2, \dots, s$).

Step 2: Decision matrix normalization. Normalize the evaluation matrix $A^{(k)} = (a_{ij}^k(\otimes))_{m \times n}$ ($k = 1, 2, \dots, s$) according to Eq. (23) to obtain the standardized evaluation matrix $\tilde{A}^{(k)} = (\tilde{a}_{ij}^k(\otimes))_{m \times n}$ ($k = 1, 2, \dots, s$).

Step 3: Calculate attribute weights ω_j^k ($j = 1, 2, \dots, n, k = 1, 2, \dots, s$), using Eqs. (24)–(34).

Step 4: Compute the comprehensive prospective values $V_i^k(\otimes)$ ($i = 1, 2, \dots, m, k = 1, 2, \dots, s$) for each alternative under each decision-maker using Eqs. (35)–(39).

Step 5: Calculate the weights $\tilde{\lambda}_k$ ($k = 1, 2, \dots, s$) for each decision-maker based on Eqs. (40)–(49).

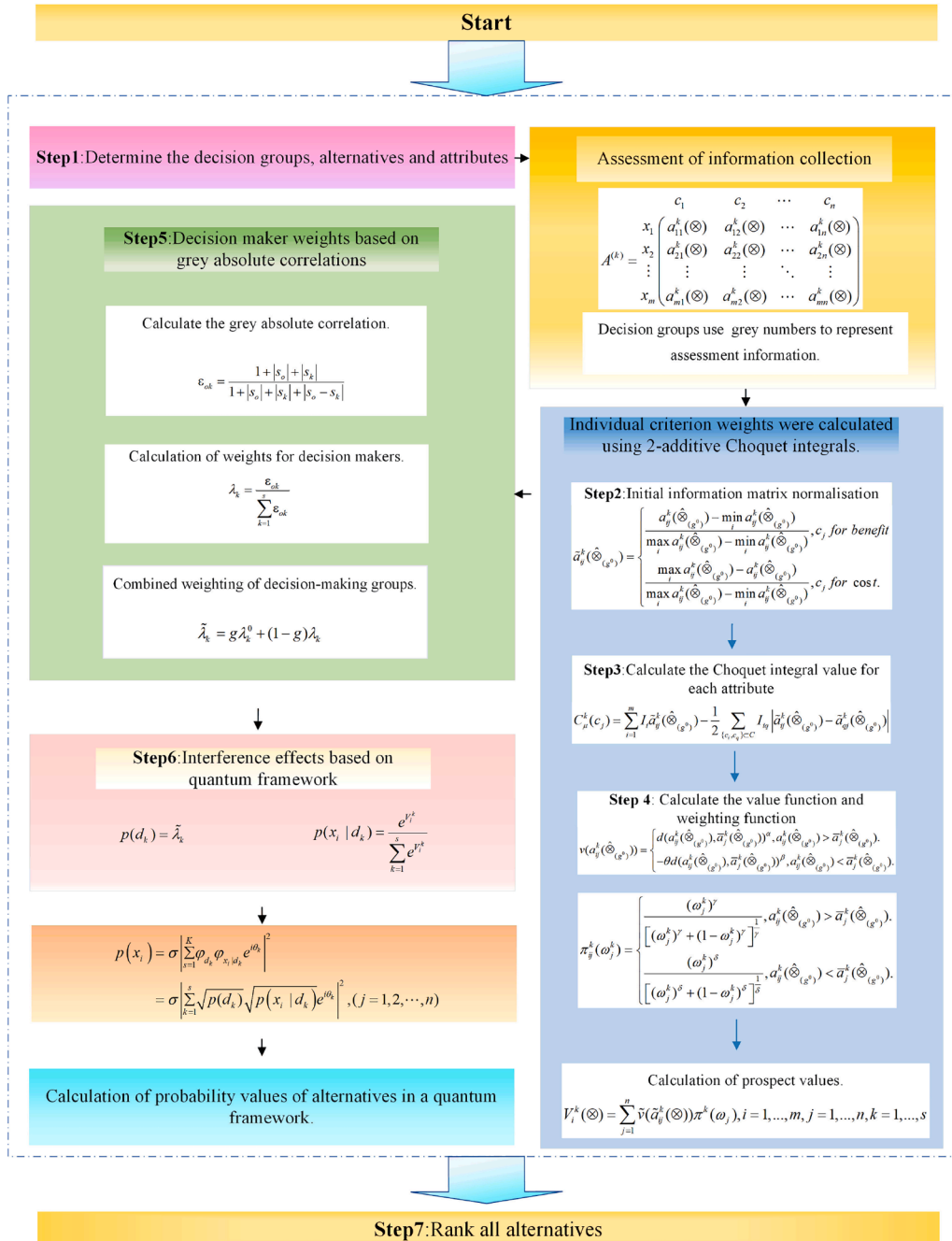


Fig. 3. Quantum group decision making steps.

Step 6: Determine the decision group's weights $\tilde{\lambda}_k (k = 1, 2, \dots, s)$ as node probabilities $p(d_k) (k = 1, 2, \dots, s)$ in the first layer of the Bayesian network by Eq. (50). Calculate conditional probabilities $p(x_i|d_k) (i = 1, 2, \dots, m, k = 1, 2, \dots, s)$ based on Eq. (51).

Step 7: The quantum probabilities $p(x_i) (i = 1, 2, \dots, m)$, for each decision alternative $x_i (i = 1, 2, \dots, m)$ are calculated by Eq. (52), and the alternatives are ranked by $p(x_i) (i = 1, 2, \dots, m)$.

4. Case analysis

In the emergency rescue of meteorological disasters, it is particularly critical to formulate and select effective emergency rescue programs. Take the devastating flood in Zhengzhou, Henan Province and its surrounding areas in July 2021 as an example. The large-scale floods, mountain torrents, mudslides and house collapses caused by the heavy rainfall have led to casualties and huge economic losses. Therefore, it is necessary to formulate an emergency rescue plan to ensure public safety and personnel evacuation. This can not only effectively reduce the losses caused by disasters, but also is the key to ensuring the safety of the lives of the affected people and promoting social stability. Assume that 5 different alternatives are set up considering different time and regional distribution such as flood discharge, personnel transfer, material delivery, and rescue force input. Consider these five attributes:

- 1) Casualties c_1 : This attribute measures the number of casualties in accidents or disasters in units of "people". This includes the count of injured and deceased individuals, with a domain of $\Omega_{c_1} \in [0, 1000]$.
- 2) Economic losses c_2 : This attribute reflects the direct economic impact of events, measured in "billion yuan". It may encompass property damage, economic activity disruptions, and other economic losses, with a domain of $\Omega_{c_2} \in [0, 1000]$.
- 3) Facility resilience c_3 : This attribute assesses the recovery capacity of facilities after experiencing a disaster, using a rating system from 0 to 10, where 0 represents complete inability to recover, and 10 represents rapid and complete recovery, with a domain of $\Omega_{c_3} \in [0, 10]$.
- 4) Flooded area c_4 : Used to describe the area affected by floods or similar disasters in units of "acres". This reflects the extent of the disaster's impact on land, with a domain of $\Omega_{c_4} \in [0, 10000]$.
- 5) Communication capability c_5 : This attribute measures the effectiveness and stability of communication systems under specific circumstances, also using a rating system from 0 to 10. A score of 0 indicates complete communication system failure, while 10 represents smooth and interference-free communication systems, with a domain of $\Omega_{c_5} \in [0, 10]$.

Three expert groups $d_k (k = 1, 2, 3)$ evaluate alternatives $x_i (i = 1, \dots, 5)$ under different attributes. The evaluation matrix is denoted as $A^{(k)} = (a_{ij}^k(\otimes))_{5 \times 5} (k = 1, 2, \dots, 5)$. Where $a_{ij}^k(\otimes)$ represents the evaluation value of expert d_k for alternative x_i under attribute c_j . Due to the uncertainty of information in emergency rescue, grey numbers are used to represent these values.

4.1. Decision-making steps

Step 1: Collect evaluation information. The evaluation information from three groups of experts are as follows.

$$A^{(1)} = \begin{bmatrix} [380, 460] & [340, 385] & [2, 5] & [5400, 5800] & [4, 6] \\ [400, 480] & [350, 390] & [3, 6] & [5200, 5600] & [5, 7] \\ [390, 450] & [330, 370] & [4, 7] & [5300, 5900] & [6, 8] \\ [370, 440] & [360, 410] & [1, 4] & [5500, 6100] & [3, 5] \\ [410, 490] & [370, 420] & [5, 8] & [5100, 5700] & [2, 4] \end{bmatrix},$$

$$A^{(2)} = \begin{bmatrix} [340, 400] & [410, 470] & [6, 9] & [5500, 6100] & [3, 4] \\ [360, 420] & [400, 490] & [7, 10] & [5300, 5900] & [2, 3] \\ [370, 430] & [430, 510] & [5, 8] & [5400, 6000] & [4, 5] \\ [330, 390] & [440, 510] & [4, 7] & [5600, 6200] & [5, 6] \\ [380, 440] & [450, 520] & [3, 6] & [5200, 5800] & [1, 2] \end{bmatrix},$$

$$A^{(3)} = \begin{bmatrix} [360, 420] & [380, 440] & [4, 7] & [5600, 6200] & [5, 8] \\ [400, 460] & [390, 450] & [3, 6] & [5300, 5900] & [6, 9] \\ [390, 450] & [400, 460] & [2, 5] & [5500, 6100] & [7, 10] \\ [370, 430] & [410, 470] & [1, 4] & [5400, 6000] & [4, 7] \\ [380, 440] & [420, 480] & [5, 8] & [5700, 6300] & [3, 6] \end{bmatrix}.$$

Step 2: Normalize each evaluation matrix $A^{(k)} = (a_{ij}^k(\otimes))_{5 \times 5} (k = 1, 2, 3)$ to obtain $\tilde{A}^{(k)} = (\tilde{a}_{ij}^k(\otimes))_{5 \times 5} (k = 1, 2, 3)$.

$$\tilde{A}^{(1)} = \begin{bmatrix} 0.6667_{(0.08)} & 0.7222_{(0.045)} & 0.25_{(0.3)} & 0.8571_{(0.04)} & 0.5_{(0.2)} \\ 0.2222_{(0.08)} & 0.4444_{(0.04)} & 0.5_{(0.3)} & 0.8571_{(0.04)} & 0.75_{(0.2)} \\ 0.6667_{(0.06)} & 1_{(0.04)} & 0.75_{(0.3)} & 0.4286_{(0.06)} & 1_{(0.2)} \\ 1_{(0.07)} & 0.2222_{(0.05)} & 0_{(0.3)} & 0_{(0.06)} & 0.25_{(0.2)} \\ 0_{(0.08)} & 0_{(0.05)} & 1_{(0.3)} & 1_{(0.06)} & 0_{(0.2)} \end{bmatrix},$$

$$\tilde{A}^{(2)} = \begin{bmatrix} 0.8_{(0.06)} & 1_{(0.06)} & 0.75_{(0.3)} & 0.25_{(0.06)} & 0.5_{(0.1)} \\ 0.4_{(0.06)} & 0_{(0.09)} & 1_{(0.3)} & 0.75_{(0.06)} & 0.25_{(0.1)} \\ 0.2_{(0.06)} & 0.3333_{(0.08)} & 0.5_{(0.3)} & 0.5_{(0.06)} & 0.75_{(0.1)} \\ 1_{(0.06)} & 0.2222_{(0.07)} & 0.25_{(0.3)} & 0_{(0.06)} & 1_{(0.1)} \\ 0_{(0.06)} & 0_{(0.07)} & 0_{(0.3)} & 1_{(0.06)} & 0_{(0.1)} \end{bmatrix},$$

$$\tilde{A}^{(3)} = \begin{bmatrix} 1_{(0.06)} & 1_{(0.06)} & 0.75_{(0.3)} & 0.25_{(0.06)} & 0.5_{(0.3)} \\ 0_{(0.06)} & 0.75_{(0.06)} & 0.5_{(0.3)} & 1_{(0.06)} & 0.75_{(0.3)} \\ 0.25_{(0.06)} & 0.5_{(0.06)} & 0.25_{(0.3)} & 0.5_{(0.06)} & 1_{(0.3)} \\ 0.75_{(0.06)} & 0.25_{(0.06)} & 0_{(0.3)} & 0.75_{(0.06)} & 0.25_{(0.3)} \\ 0.5_{(0.06)} & 0_{(0.06)} & 1_{(0.3)} & 0_{(0.06)} & 0_{(0.3)} \end{bmatrix}.$$

Step 3: Calculate the 2-additive Choquet integral values for each attribute. The significance of each attribute specified in the classification process may be determined by doing calculations using Eqs. (24)–(29). Then, calculate the Shapley value for individual characteristics and the interaction values for attributes using Eqs. (30)–(32). Finally, utilize Eq. (33) to compute the integral values for each attribute, and the results are as follows.

$$C_u^1(C) = (0.0497, 0.0833, 0.034, 0.1674, 0.1146),$$

$$C_u^2(C) = (0.0030, 0.0190, 0.1327, 0.0174, 0.0215),$$

$$C_u^3(C) = (0.1515, 0.1483, 0.0382, 0.0889, 0.0975).$$

Step 4: To calculate the prospective values of each alternative, prospect theory is employed. This is achieved by utilizing the equations from Eqs. (35)–(39).

$$V^1(X) = (-0.1978, -0.2844, -0.0953, -0.1933, -0.3974)^T,$$

$$V^2(X) = (-0.0328, -0.0816, 0.0942, -0.1539, -0.8104)^T,$$

$$V^3(X) = (-0.5397, -0.2054, 0.1110, -0.5002, 0.0691)^T.$$

Step 5: Based on grey absolute correlation, assuming the subjective weights of the decision group are $\lambda^0 = (0.5, 0.3, 0.2)^T$, calculate the decision-maker's weight model by applying Eqs. (40)–(49).

$$\tilde{\lambda}_1 = 0.4122, \quad \tilde{\lambda}_2 = 0.3160, \quad \tilde{\lambda}_3 = 0.2718.$$

Step 6: Define the content and structure of the Bayesian network. Converting the decision group's weights into probabilities within the Bayesian network, we obtain the following:

$$p(d_1) = 0.4122, \quad p(d_2) = 0.3160, \quad p(d_3) = 0.2718.$$

Based on the prospective values of each attribute, the conditional probabilities within the Bayesian network can be calculated using

Eq. (51) as follows:

$$\begin{aligned} p(x_1|d_1) &= 0.2063, p(x_2|d_1) = 0.1891, p(x_3|d_1) = 0.2285, \\ p(x_4|d_1) &= 0.2072, p(x_5|d_1) = 0.1689, p(x_1|d_2) = 0.2256, \\ p(x_2|d_2) &= 0.2148, p(x_3|d_2) = 0.2561, p(x_4|d_2) = 0.1998, \\ p(x_5|d_2) &= 0.1036, p(x_1|d_3) = 0.1390, p(x_2|d_3) = 0.1942, \\ p(x_3|d_3) &= 0.2665, p(x_4|d_3) = 0.1446, p(x_5|d_3) = 0.2556. \end{aligned}$$

Step 7: Utilize Eq. (52) to obtain the quantum probabilities for each alternative:

$$\begin{aligned} p(x_1) &= \left| \sum_{k=1}^3 \sqrt{p(d_k)} \sqrt{p(x_1|d_k)} e^{i\Theta_k} \right| \\ &= 0.1941 + 0.1557 \cos \Theta_{12} + 0.1134 \cos \Theta_{13} + 0.1038 \cos \Theta_{23}, \end{aligned}$$

$$\begin{aligned} p(x_2) &= \left| \sum_{k=1}^3 \sqrt{p(d_k)} \sqrt{p(x_2|d_k)} e^{i\Theta_k} \right| \\ &= 0.1986 + 0.1455 \cos \Theta_{12} + 0.1283 \cos \Theta_{13} + 0.1197 \cos \Theta_{23}, \end{aligned}$$

$$\begin{aligned} p(x_3) &= \left| \sum_{k=1}^3 \sqrt{p(d_k)} \sqrt{p(x_3|d_k)} e^{i\Theta_k} \right| \\ &= 0.2476 + 0.1746 \cos \Theta_{12} + 0.1652 \cos \Theta_{13} + 0.1531 \cos \Theta_{23}, \end{aligned}$$

$$\begin{aligned} p(x_4) &= \left| \sum_{k=1}^3 \sqrt{p(d_k)} \sqrt{p(x_4|d_k)} e^{i\Theta_k} \right| \\ &= 0.1879 + 0.1469 \cos \Theta_{12} + 0.1159 \cos \Theta_{13} + 0.0996 \cos \Theta_{23}, \end{aligned}$$

$$\begin{aligned} p(x_5) &= \left| \sum_{k=1}^3 \sqrt{p(d_k)} \sqrt{p(x_5|d_k)} e^{i\Theta_k} \right| \\ &= 0.1718 + 0.0955 \cos \Theta_{12} + 0.1391 \cos \Theta_{13} + 0.0954 \cos \Theta_{23}. \end{aligned}$$

The QLBN proposed in this paper constructs quantum probabilities by integrating decision-maker weights (as prior probabilities) and foreground-based conditional probabilities. Interference between decision makers is modeled by the phase angle, which controls the amplitude and the cosin-based interference term. The quantum probability of each option consists of two parts: the classical weighted term and the quantum interference term. When all phase angles are 90° , the interference term disappears and the model degenerates into a classical weighted aggregate model, indicating that the method allows behavior-level expansion under cognitive interaction while maintaining compatibility with traditional frameworks. Therefore, this paper ranks the alternatives according to quantum probability. Assuming there is no interference effect between decision groups d_1 and d_3 , and between d_2 and d_3 . Only the interference effect between d_1 and d_2 is considered, so different phase angle angles have been set to obtain the probability values of alternative alternatives.

This paper provides the variation of probability scores as Θ_{12} changes from 0 to 2π , as shown in Fig. 4. From the figure, it can be observed that the graph is symmetric about $\Theta_{12} = \pi$, so the paper only need to focus on $\Theta_{12} \in [0, \pi]$ (Wu et al., 2021, 2022). The specific analysis is as follows:

- 1) When $\Theta_{12} = \frac{\pi}{2}$, there is no interference between d_1 and d_2 , as they are completely independent of each other. Thus, in this scenario, the outcomes align with those of the traditional weighted average fusion model, resulting in the final ranking order of $x_3 > x_1 > x_2 > x_4 > x_5$.
- 2) When $\Theta_{12} \in [0, \frac{\pi}{2}]$, and there is constructive interference between d_1 and d_2 . Despite the drop in results, the positive influence still leads to greater outcomes compared to the independent scenario. The final ranking order is $x_3 > x_2 > x_1 > x_4 > x_5$.

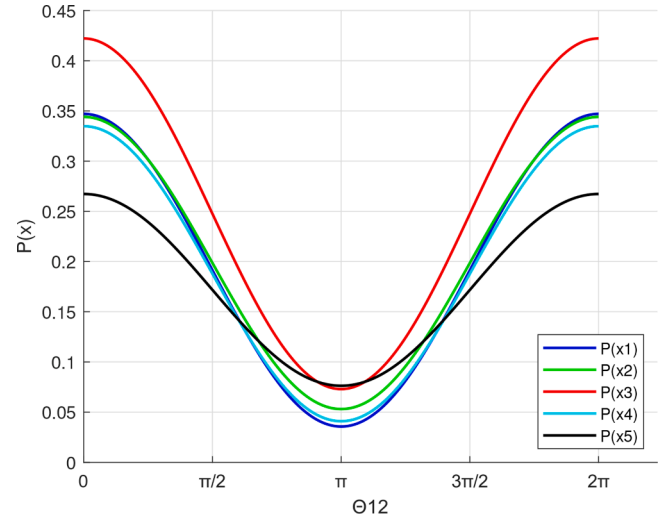


Fig. 4. Probabilities of alternatives along Θ_{12} changes.

- 3) When $\Theta_{12} \in [\frac{\pi}{2}, \pi]$, there is negative interference between d_1 and d_2 . On one hand, the results decrease, and on the other hand, the negative effect leads to results lower than in the independent case. The final ranking order is $x_5 > x_3 > x_2 > x_4 > x_1$.

On this basis, in order to make the final recommendation ranking reflect the balance between group preference and structural robustness. In this paper, positive interference conditions are selected as the final recommended ranking for the following reasons: 1) Under neutral and positive interference conditions, alternative x_3 is always the optimal result, showing stability, showing stability; 2) Positive interference reflects group dynamics common in the real world, where members tend to reach consensus rather than confrontation; 3) The order structure obtained is smooth without sudden inversion, indicating that the model is structurally robust under positive interaction; 4) Compared with the negative disturbance order that shows drastic changes, the positive disturbance condition is more interpretable and acceptable in behavior. Therefore, the final alternative ranking result is: $x_3 > x_1 > x_2 > x_4 > x_5$. In this alternative, the optimal solution is determined as x_3 , indicating that it is the preferred choice after comprehensive assessment as it best satisfies multiple criteria, handles interference effects, and aligns with decision objectives.

4.2. Sensitivity analysis

In this section of the paper, the attention is directed towards examining how the parameter g in the model influences the final calculation and ranking results. A critical role is played by this parameter within the model, representing the balancing coefficient when subjective and objective weights are assigned by decision groups.

As the parameter value increases, a tendency is observed in the model to assign more weight to the subjective component. This implies that a more critical role is played by decision-makers' individual preferences in the final decision-making process. The range of values for this parameter is set from 0 to 1, incremented by 0.2 at each interval. This configuration allows for the observation of how changes in weight allocation, influenced by parameter values, impact the final decision results, transitioning from entirely objective to entirely subjective.

Additionally, the setting of phase angles is also encompassed in the research, with values set at 0, $\frac{1}{2}\pi$, and 2π . The use of phase angles in this context is for simulating variations in decision-maker attitudes towards different alternatives. By varying both parameter values and phase angles, decision outcomes under different combinations can be observed by researchers.

Table 2
Probabilities of alternatives with various Θ_{12} and g .

Θ_{12}	g	$p(x_1)$	$p(x_2)$	$p(x_3)$	$p(x_4)$	$p(x_5)$	Ranking order
0	0	0.3312	0.3317	0.4095	0.3168	0.2639	$x_3 > x_2 > x_1 > x_4 > x_5$
	0.2	0.3390	0.3370	0.4150	0.3243	0.2655	$x_3 > x_1 > x_2 > x_4 > x_5$
	0.4	0.3463	0.3419	0.4199	0.3314	0.2668	$x_3 > x_1 > x_2 > x_4 > x_5$
	0.6	0.3552	0.3463	0.4243	0.3380	0.2678	$x_3 > x_1 > x_2 > x_4 > x_5$
	0.8	0.3596	0.3503	0.4283	0.3442	0.2686	$x_3 > x_1 > x_2 > x_4 > x_5$
$\frac{\pi}{2}$	1	0.3657	0.3540	0.4318	0.3501	0.2692	$x_3 > x_1 > x_2 > x_4 > x_5$
	0	0.1896	0.1994	0.2507	0.1833	0.1770	$x_3 > x_2 > x_1 > x_4 > x_5$
	0.2	0.1914	0.1991	0.2495	0.1851	0.1750	$x_3 > x_2 > x_1 > x_4 > x_5$
	0.4	0.1932	0.1988	0.2482	0.1869	0.1729	$x_3 > x_2 > x_1 > x_4 > x_5$
	0.6	0.1950	0.1985	0.2469	0.1888	0.1708	$x_3 > x_2 > x_1 > x_4 > x_5$
π	0.8	0.1968	0.1982	0.2457	0.1906	0.1687	$x_3 > x_2 > x_1 > x_4 > x_5$
	1	0.1986	0.1979	0.2444	0.1925	0.1667	$x_3 > x_2 > x_1 > x_4 > x_5$
1-12	0.1	0.0480	0.0671	0.0919	0.0497	0.0902	$x_3 > x_5 > x_2 > x_4 > x_1$
	0.2	0.0438	0.0612	0.0839	0.0459	0.0844	$x_5 > x_3 > x_2 > x_4 > x_1$
	0.4	0.0401	0.0557	0.0765	0.0425	0.0790	$x_5 > x_3 > x_2 > x_4 > x_1$
	0.6	0.0368	0.0507	0.0695	0.0396	0.0738	$x_5 > x_3 > x_2 > x_4 > x_1$
	0.8	0.0340	0.0460	0.0631	0.0370	0.0689	$x_5 > x_3 > x_2 > x_4 > x_1$
π	1	0.0315	0.0417	0.0570	0.0349	0.0642	$x_5 > x_3 > x_2 > x_4 > x_1$

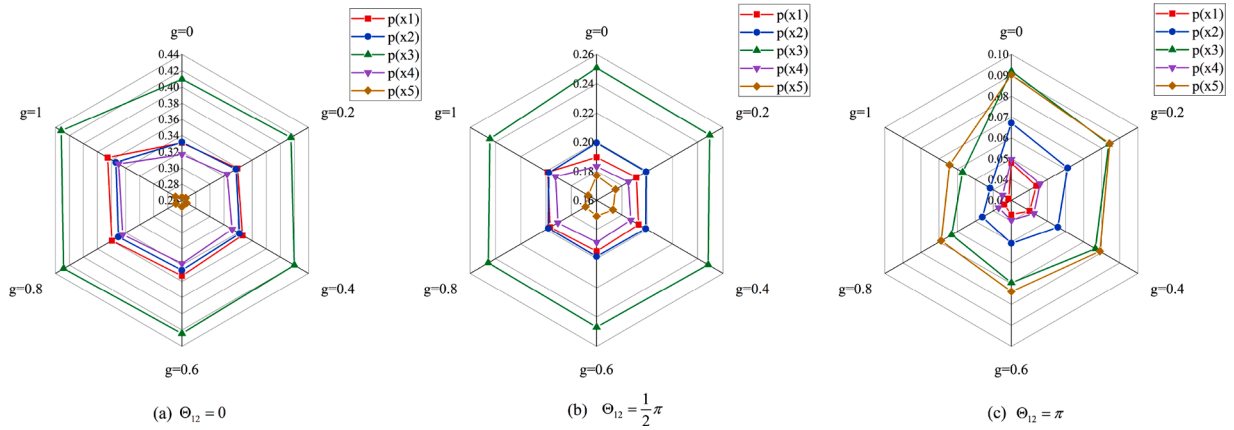


Fig. 5. Probabilities of alternatives along Θ_{12} and weighting coefficient g change.

This method facilitates a more profound comprehension of the interplay between subjective and objective aspects in the process of group decision-making and their impact on the ultimate rankings and choices. By adjusting the parameters and making observations, we achieved findings that are shown in Table 2 and Fig. 5. These results help improve and optimise the MAGDM model to better suit different decision environments and requirements.

Through the analysis of calculations related to Θ_{12} and weighting coefficient g , distinct ranking orders have been discerned. The primary findings of this paper encompass:

- 1) When the values of Θ_{12} are constant, changes in the values of g cause alterations in the probability scores for every alternative, but the overall rankings generally stay the same. Table 2 provides comprehensive data on the ranking of alternative options for various Θ_{12} and g values. The oscillations in Θ_{12} and g are graphically shown in Fig. 5, illustrating the patterns in the probability scores of various different options.
- 2) The changes in probability scores of alternative alternatives exhibit a fluctuating distribution consistent with the characteristics of amplitude variation.
- 3) The phase angle Θ_{12} significantly impacts the ranking of alternative alternatives. For instance, when interference factors positively affect the alternative, x_1 performs optimally, whereas in situations where interference factors have a negative impact, the alternative x_5 emerges as the optimal choice.

Therefore, this paper discovered that a particular coefficient g has a negligible impact on the ranking order, suggesting its lack of sensitivity to this parameter. Conversely, the phase angle Θ_{12} showed considerable sensitivity to the order of ranking. The results highlight the high level of efficiency and efficacy of the GMAQGDGM suggested in this study. The model demonstrates exceptional proficiency in improving flexibility and stability in practical decision-making options.

Subsequently, this paper delves into the examination of mutual interference effects among different decision groups in calculations and final rankings. Interference effects between d_1 and d_3 , as well as between d_2 and d_3 , are taken into account, while d_1 and d_2 are treated as independent entities. Through the manipulation of various phase angles $\Theta_{13} = \gamma$, $\Theta_{23} = \beta$, the research found that the probability scores of alternative alternatives exhibit symmetric and dynamic characteristics, following some interpretable patterns. These results are presented in Fig. 6.

To clearly demonstrate the impact of interference on probability scores, the 3D data model is projected onto a 2D plane. This transformation leverages specialized data visualization techniques to facilitate a more intuitive understanding of data patterns and trends. The outcomes depicted in Fig. 7 unveil the fluctuations in probability scores under diverse parameters, delineating the intricacy of interactions between these parameters. Moreover, this 2D visualization method accentuates critical features in the data, offering a fresh perspective for data analysis. These observations culminate in several pivotal conclusions, encompassing critical trends in the data model and the

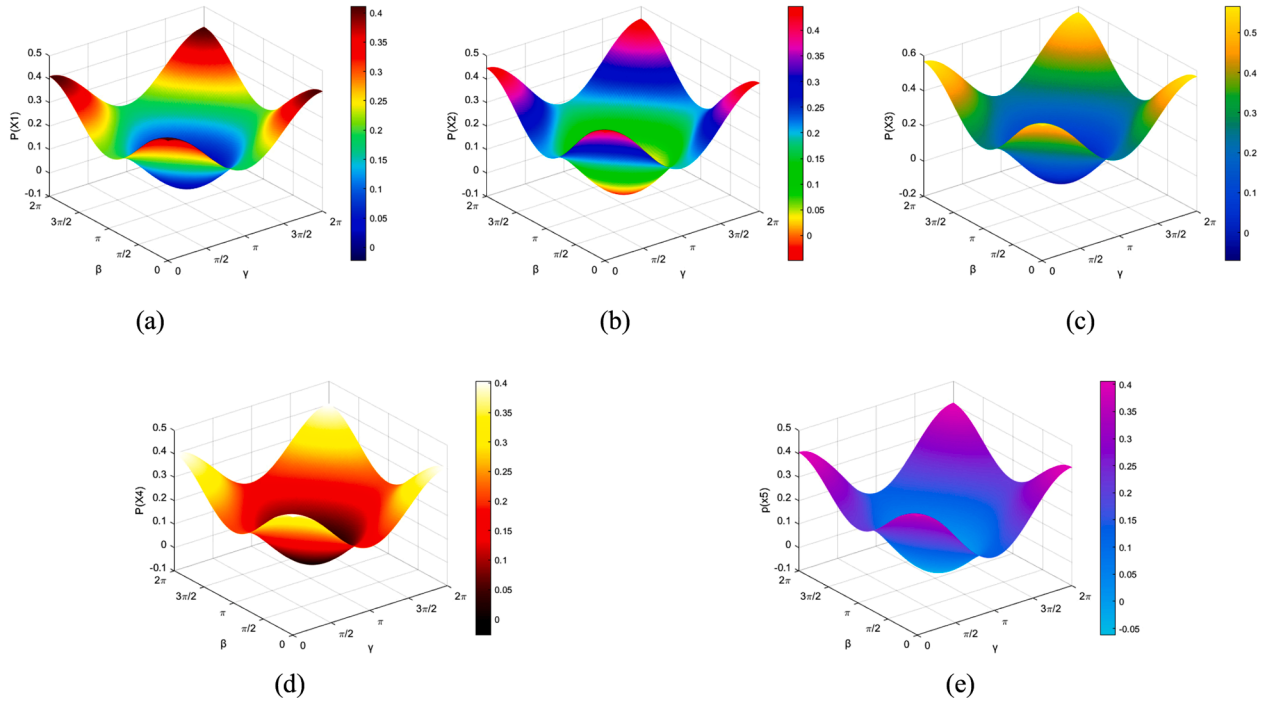


Fig. 6. Probabilities of alternatives along γ and β change.

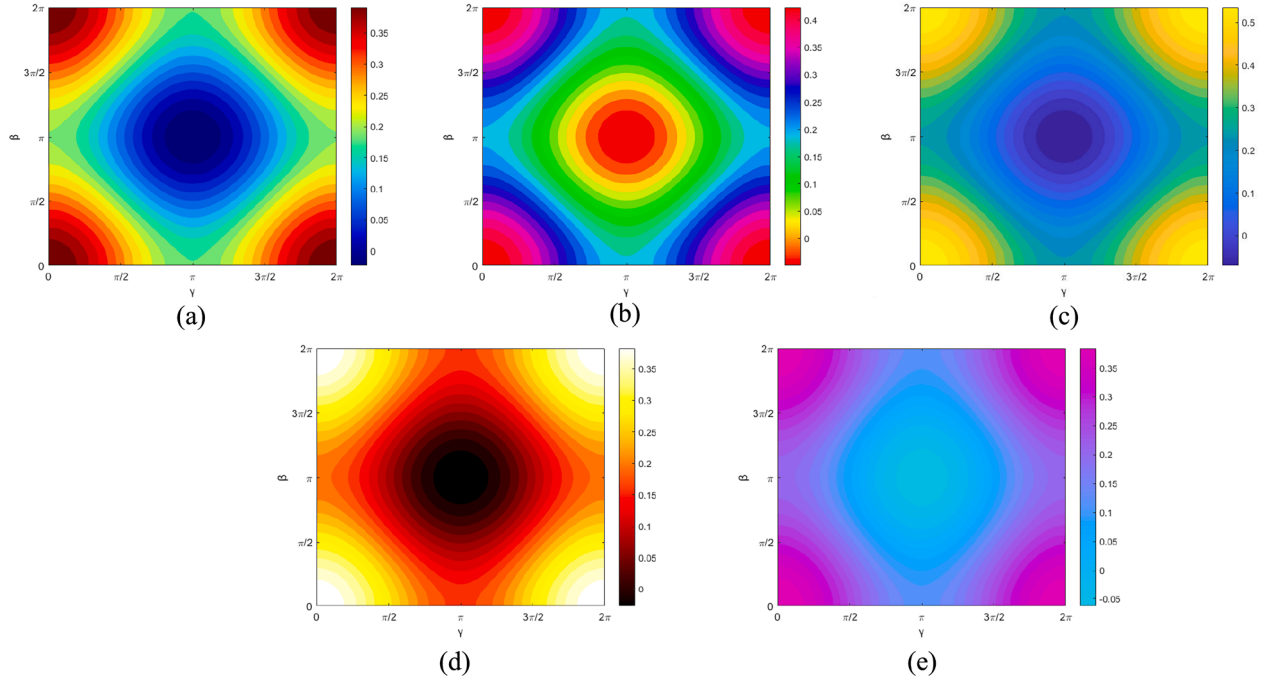


Fig. 7. Probabilities of alternatives along γ and β change.

ramifications of interactions between parameters. The conclusions are outlined as follows:

- 1) Symmetry is observed in the probability scores of alternatives concerning (π, π) . Therefore, while studying interference effects, it is satisfactory to analyse the interval of $[0, \pi] \times [0, \pi]$. This implies that when studying interference effects, it is sufficient to analyse data on one side of the symmetric axis in order to have a thorough knowledge of these effects. This approach simplifies the analytical process.
- 2) The relationship between the probability scores of alternate options and the phase angles β and γ demonstrates nonlinearity. The calcula-

tion results manifest fluctuating changes, providing a more nuanced representation of interference effects. Nonlinearity indicates the lack of a simple, direct correlation between probability scores and phase shifts. Instead, more intricate interactions give rise to fluctuating changes in probability scores, offering a more accurate reflection of the complexity inherent in interference effects.

- 3) Within the phase range of $[0, \pi] \times [0, \pi]$, an increment in phase results in more negative interference effects among any two choice groups. Consequently, lower probability scores ensue, causing diminished results. Under specific conditions, heightened phase values exacerbate negative interference effects between decision groups, impacting

Table 3
Comparison of different methods.

Methods	Ranking indices	Final ranking orders
TOPSIS (Susmaga, Szczęch, & Brzezinski, 2024)	$C_1^* = 0.7271, C_2^* = 0.5048,$ $C_3^* = 0.6781, C_4^* = 0.4647,$ $C_5^* = 0.2568.$	$x_1 > x_3 > x_2 > x_4 > x_5$
TODIM (Llamazares, 2018)	$\phi(x_1) = 0.8602, \phi(x_2) = 0.5569,$ $\phi(x_3) = 0.6297, \phi(x_4) = -0.8644,$ $\phi(x_5) = -1.1824.$	$x_1 > x_3 > x_2 > x_4 > x_5$
Quantum framework of (Wu et al., 2021)	$p(x_1) = 0.5601, p(x_2) = 0.3209,$ $p(x_3) = 0.5353, p(x_4) = 0.1608,$ $p(x_5) = 0.0780.$	$x_1 > x_3 > x_2 > x_4 > x_5$
Quantum framework of (Wu et al., 2022)	$p(x_1) = 0.3498, p(x_2) = 0.3441,$ $p(x_3) = 0.4222, p(x_4) = 0.3347,$ $p(x_5) = 0.2674.$	$x_3 > x_2 > x_1 > x_4 > x_5$
The proposed method	$p(x_1) = 0.3039, p(x_2) = 0.3122,$ $p(x_3) = 0.3248, p(x_4) = 0.2888,$ $p(x_5) = 0.3002.$	$x_3 > x_1 > x_2 > x_4 > x_5$

- final decision outcomes and resulting in decreased probability scores—an indication of reduced likelihood for those results. This finding holds significance for a comprehensive understanding and effective management of interference impacts in decision-making processes.
- The point $\left[\frac{\pi}{2}, \frac{\pi}{2}\right]$ inside the specified range establishes a crucial boundary. At this juncture, decision groups operate independently without interference, indicating that the behaviour and choices of different decision groups do not mutually influence each other. However, deviating from this point initiates the manifestation of interference effects, impacting the decision-making processes of each decision group. This interference results in notable disparities in probability scores from the ideal state at point $\left[\frac{\pi}{2}, \frac{\pi}{2}\right]$, even among decision groups with some similarity. Therefore, recognizing and comprehending this critical point is imperative for optimizing the decision-making process, particularly when assessing and managing interactions among multiple decision groups.
 - Within the range of $\left[\frac{\pi}{2}, \frac{\pi}{2}\right] \times \left[\frac{\pi}{2}, \frac{\pi}{2}\right]$, positive interference effects between β and γ yield a disjunctive result. This indicates that, during their interaction, these factors incline the decision process towards choosing one or the other. On the other hand, in a different range $\left[\frac{\pi}{2}, \pi\right] \times \left[\frac{\pi}{2}, \pi\right]$, these factors produce detrimental interference effects, leading to a combined result. In this situation, the choice takes into account both criteria at the same time instead than showing preference for one over the other. Interactions between components in various ranges may result in results that include features of both disjunction and conjunction, or more complex choice patterns. The existence of such interference effects renders the decision-making process more intricate and nuanced, necessitating consideration of interactions between different factors and how these interactions evolve under varying conditions.

4.3. Comparative analysis

4.3.1. Comparative analysis of alternative ranking results

In order to assess the efficacy of the strategy proposed in this paper, a comparison study was performed, comparing it with several current decision-making models. These comparative models may be directly used or indirectly modified to tackle the case issue presented in this paper. The final ranking results obtained from various approaches are shown in Table 3.

In comparison with traditional multi-attribute decision-making methods, the core idea of TOPSIS is to sort alternatives by calculating the distance between them and the positive ideal solution and the negative ideal solution, and select the alternative closest to the ideal solution as the optimal solution. In this example, TOPSIS (Susmaga et al., 2024) method determines alternative x_1 as the optimal choice, which

is different from this paper. However, based on the weighted average of attributes, this method ignores the possible interaction between attributes, and does not consider the subjective interference and cognitive influence between decision makers, which is easy to introduce structural bias in group integration. Similarly, although the TODIM (Llamazares, 2018) method introduces a relative advantage function based on loss avoidance to characterize part of psychological preference behavior, its essence is still an aggregation method based on individual summation, and no interaction modeling mechanism between decision makers has been established. In the same case, the TODIM method also regards x_1 as the optimal alternative, and the sorting result is significantly different from that of the proposed method, indicating that the traditional method has obvious limitations in dealing with interference effects and attribute coupling in group decision making.

Further, the paper compares with two typical quantum group decision models. The quantum model in Quantum model 1 (Wu et al., 2021) introduces the interference modeling mechanism, which can theoretically reflect the interference effect of preference among individuals, but its attribute weight and decision-makers' weights are completely dependent on subjective assignment and lack of objective basis. In this model, the optimal alternative is still x_1 , which is obviously different from the output of the model in this paper. On this basis, the Quantum model 2 (Wu et al., 2022) introduces the entropy weight method to calculate the attribute weight, and determines the decision-maker's weight through the similarity measure, which improves the objectivity and data dependence of weight calculation. Under this model, the optimal alternative is the same as x_3 in this paper, but there are still differences in the ordering of other alternative schemes. This is mainly due to the fact that the model does not consider the possible nonlinear interaction between attributes, and is relatively simple in the modeling of interference structure, which cannot fully capture the multi-level influence mechanism among decision makers.

However, this paper first considers the interaction between attributes, and constructs a 2-additive Choquet integral model based on Matten system to solve the optimal attribute weights. Secondly, this paper puts forward a weight fusion mechanism that combines the absolute grey correlation degree and subjective weighting to achieve a unified expression of the rationality of decision-makers' weight and individual differences. Finally, the prospect theory is introduced into conditional probability modeling of quantum Bayes structure to simulate the interaction between decision makers' risk preference and individuals. Therefore, the method proposed in this paper is more suitable for uncertain complex decision scenarios and has better robustness and robustness.

4.3.2. Comparative analysis of weight models

In order to further verify the stability and robustness of the proposed decision maker weight determination method under the decision background of uncertainty and complexity, a systematic comparison

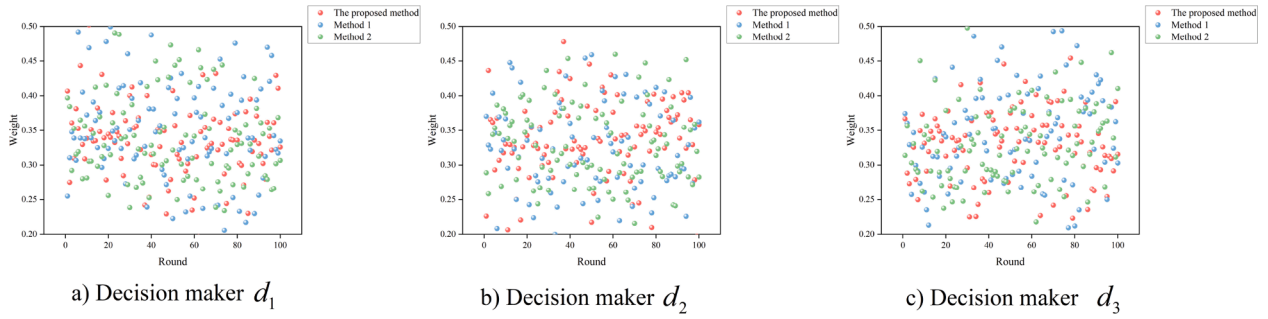


Fig. 8. Comparison of weight distribution of different decision makers.

experiment is designed to compare the proposed method with traditional methods. The experimental setting consisted of three decision makers, and the evaluation information of each decision maker was represented as a 5×5 matrix. The evaluation data were randomly generated by MATLAB's rand function within the interval $[0,1]$ to simulate the performance of group preferences in decision situations with incomplete information, strong subjectivity and significant data noise. In view of the differences in knowledge, experience and other aspects of decision makers in different decision-making situations, the weight of decision makers is also different, so this experiment assumes that the subjective weight of each decision maker is also randomly generated to simulate the change of weight of different decision makers in diverse decision-making scenarios. On this basis, the model proposed in this paper is compared with two representative weight determination models: the entropy-based method 1 (Yang, Pan, & Zeng, 2022) and the similarity-based method 2 (Wang, Xu, & Xu, 2019), and the process is repeated 100 times to systematically observe the fluctuation characteristics of decision-maker weights output by the three methods under different random input conditions, as shown in Fig. 8.

Fig. 8 shows the weight changes of three decision makers over 100 rounds of simulation experiments. Judging from the overall trend, the weight output distribution of the proposed method on the three decision makers is obviously more stable, with small fluctuation range and high concentration, which reflects good numerical stability and anti-interference ability. In contrast, Method 1 (Yang et al., 2022) measures the contribution of decision makers' information based on entropy distribution. However, entropy is extremely sensitive to local extreme values, resulting in a large shift in output weights when the input data is slightly disturbed, showing obvious instability. Although consistency is used as the weight adjustment index in Method 2 (Wang et al., 2019), it relies more on the consistent trend among evaluation information. Under the condition of strong evaluation heterogeneity and lack of a prior consensus structure, the output is easily disturbed by noise and produces violent fluctuations, especially in the weight distribution of d1 and d3, which shows obvious dispersion.

However, by introducing grey relational degree to describe the structural similarity between individual preference and group cognition, and integrating the subjective weight fusion mechanism, this method has structural advantages in dealing with information incompleteness and cognitive bias. This fusion strategy not only enhances the model's explanatory ability and behavioral adaptability in highly uncertain environments, but also improves the robustness and consistency of group members' importance recognition. In summary, the weight method proposed in this paper shows strong convergence and stability under multiple rounds of stochastic simulation, and has good robustness.

4.4. Simulation analysis

In order to verify the stability and robustness of the decision maker weight calculation method proposed in this paper, simulation experiments are carried out. A total of 3 groups of 5×5 decision matrices were

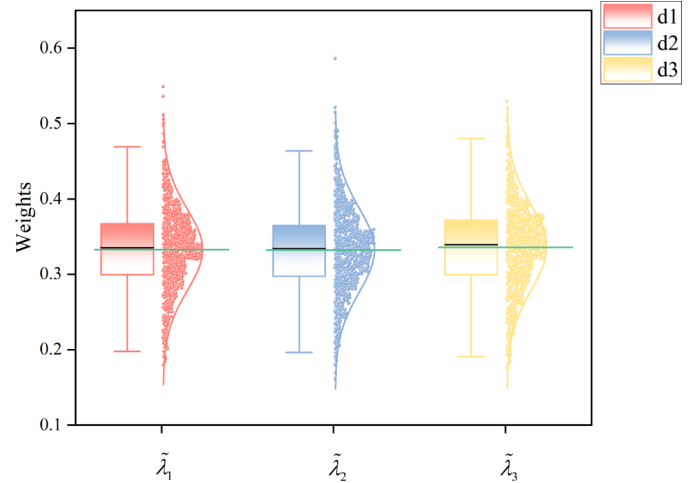


Fig. 9. The simulation analysis of the decision maker weight model.

constructed in the experiment, in which the evaluation values of each decision maker on different attributes of each alternative scheme were randomly generated by the rand function in MATLAB to simulate the input of decision preferences with uncertainty and heterogeneity in real scenarios. The method for determining the weight of the decision maker in this paper is run independently on each set of data, and the process is repeated 1000 times to systematically evaluate the output stability of the method under input perturbations, as shown in Fig. 9.

Fig. 9 shows the weight distribution results of the three decision makers in 1000 experiments, visualized using a boxplot combined with the normal kernel density estimate curve (right). The green horizontal line represents the mean position of the weights, the box represents the median and quartile range, and the whisker line represents the extreme value range. It can be observed from the results that the weight mean and median of the three decision makers almost coincide, indicating that the weight distribution is highly symmetrical and concentrated, reflecting the stable output characteristics of the method under multiple iterations. At the same time, the shape of the kernel density curve is close to normal distribution, and the distribution fluctuation is small, which indicates that the method has good robustness and normal convergence in the face of random disturbance.

5. Conclusion

In order to deal with multi-source heterogeneous information, attribute interaction and group cognitive interference more effectively in the context of complex uncertain decision making, this paper proposes a multi-attribute grey quantum group decision making method that integrates grey number, 2-Choquet integral and quantum probability wheel. The approach builds a unified modeling framework for information fusion and preference evolution, with three key contributions:

- 1) 2-additive Choquet integral method based on MTS is used to calculate attribute weights. With the support of non-additive fuzzy measure theory, this method models the first-order and second-order interactions between attributes of the system, breaks through the traditional attribute independence assumption, and depicts the cooperative relationship between multiple attributes more reasonably.
- 2) The weight fusion mechanism of subjective and objective decision makers based on grey relational degree. The absolute relational degree between individual preference and group preference is calculated by grey system theory, and objective weight is obtained, then the comprehensive weight with both group representation and individual difference expression is obtained by fusion with subjective empowerment information.
- 3) A quantum-like Bayesian inference framework considering the interference effect of decision makers. In this framework, the preference information under finite rationality is mapped into conditional probability vector by prospect theory, and quantum interference structure is constructed by combining prior probability, and then the quantum probability value of each scheme is calculated to realize the scheme ranking and optimization.

To sum up, the grey quantum group decision-making method proposed in this paper achieves an organic integration in key links such as information representation, weight modeling and interference reasoning, and constructs a unified decision analysis framework. This method has good applicability and robustness for dealing with group decision-making problems with high uncertainty.

However, the approach described in this paper requires prior identification of any interference among decision makers. Furthermore, the decision outcomes for each case must be examined in light of both positive and negative interference among decision makers. As the number of decision makers grows, the computational workload increases significantly. In order to tackle this difficulty, the rising complexity will be managed by the use of machine learning methods. Subsequent studies will investigate the incorporation of machine learning into the quantum group decision-making process, specifically by identifying the interference values among decision makers and using the quantum framework in group decision-making situations on a wide scale.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and we do not have any possible conflicts of interest.

CRediT authorship contribution statement

Shuli Yan: Writing – original draft, Methodology, Conceptualization; **Yizhao Xu:** Writing – original draft, Writing – review & editing, Visualization, Methodology; **Francisco Javier Cabrerizo:** Writing – review & editing, Supervision; **Enrique Herrera-Viedma:** Conceptualization, Writing – review & editing, Supervision.

Acknowledgment

This work is funded by the [National Natural Science Foundation of China \(72471124, 72371137, 71801085\)](#), the Social Science Fund Project of Jiangsu Province (23GLB006), Jiangsu University Philosophy and Social Science project (2022SJYB0187), the National Scholarship Foundation of China (202209045002), PID2022-139297OB-I00 funded by MICIU/AEI/10.13039/501100011033 and by ERDF/EU, and by the

project C-ING-165-UGR23, co-funded by the Regional Ministry of University, Research and Innovation and by the European Union under the Andalusia ERDF Programme 2021–2027.

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